Delivery Delay & Logistics Analysis

Data-Driven Insights for Delivery Reliability & Customer Trust

Prepared by: Sanket Nagdive
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Project Overview & The "Trust Gap"

The Business Problem

Estimated delivery dates are frequently over-conservative — often twice the actual delivery time. This causes potential customers to abandon purchases before checkout.

The Trust Gap

When carriers genuinely run late, customers lose trust. The dual problem of over-promising and under-delivering creates a "Trust Gap" that impacts retention.

Our Goal

Identify root causes of delivery delays and improve estimation accuracy using machine learning.

"How can we provide a more accurate — and shorter — delivery estimate to increase conversion without over-promising?"

Data Architecture & Tech Stack

Data Sources

- Orders & Order Items
- Payments & Products
- Sellers & Geolocation
- MySQL Database

Tech Stack

- Python (Pandas, SQLAlchemy)
- SQL (Complex Joins & CTEs)
- Power BI Dashboards
- Logging & Execution Tracking

Process

- Automated CSV Ingestion
- Database Schema Setup
- Feature Engineering in SQL
- Exploratory Data Analysis

Feature Engineering for Logistics

TIME-BASED

Order Handling Days

Measures the efficiency of the internal approval process from purchase to approval.

```
DATEDIFF(approved_at, purchase_timestamp)
```

LOGISTICS

Approval Days

Tracks the duration from order approval to the actual handoff to the carrier.

```
DATEDIFF(carrier_date, approved_at)
```

CUSTOMER

Total Delivery Days

The end-to-end duration from the customer's perspective (Purchase to Delivery).

```
DATEDIFF(customer_date, purchase_timestamp)
```

PERFORMANCE

Delay Flags

Binary indicators to identify orders that exceeded estimated or shipping limits.

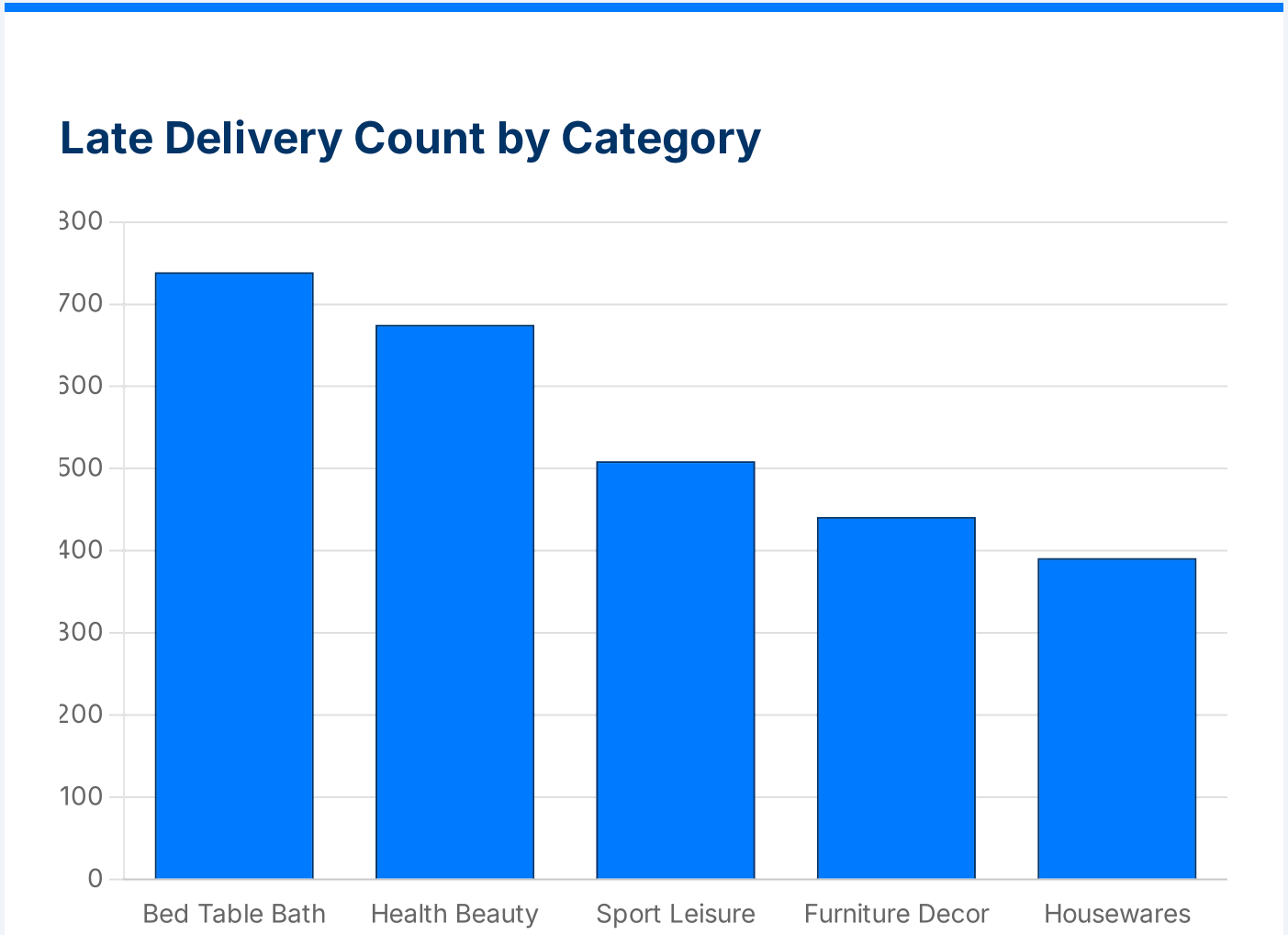
```
CASE WHEN delivery > estimated THEN 1 ELSE 0
```

Late Delivery Deep Dive - Categories

Top 5 Problematic Categories

Analysis reveals that specific product categories suffer significantly more from delivery failures than others.

Bulky, high-volume, and potentially fragile categories like "Bed Table Bath" and "Health Beauty" are most susceptible to carrier and handling delays.



Seller Performance Analysis

Seller ID (Truncated)	Late Deliveries
7c67e1448b...	200
4a3ca9d8b3...	118
1f50f92017...	115
6560211a19...	108
cc419e0650...	105

Key Findings

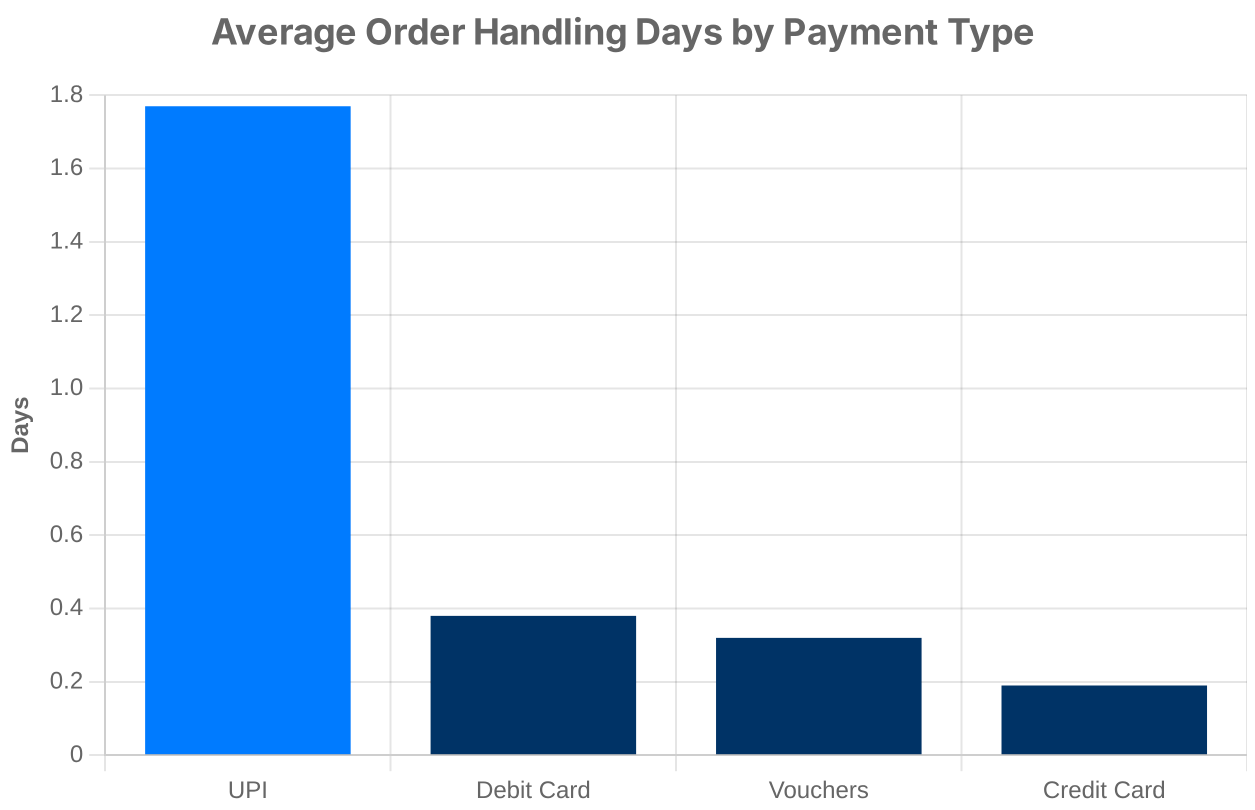
The top seller responsible for late deliveries shows **200 late orders** — nearly double the next highest seller.

A small group of sellers contributes disproportionately to overall delivery failures.

Action Plan

Implement performance improvement plans and **SLA enforcement** with financial penalties for persistent lateness.

Payment Bottlenecks: The UPI Delay



The 9x Gap

UPI payments show a significantly higher average handling time of **1.77 days** compared to credit cards at **0.19 days**.

Root Cause Analysis

This suggests UPI payment confirmation latency is a major bottleneck in the order approval pipeline, likely due to integration or banking API delays.

Correlation: Weight, Freight & Speed

Source: Delivery Performance Dataset Analysis

Weight Correlation

A clear positive correlation exists: heavier products with higher freight charges take significantly longer to deliver.

Key Comparison

Low Freight (~581g): **5.7 Days**

Very High Freight (~5.6kg): **15.2 Days**

ML Insight

Product weight and freight value are the strongest predictors of delivery time, making them essential features for the ML estimator.

Power BI Dashboard Overview

Interactive Reporting

Stakeholders can dynamically filter logistics data by product category, top seller ID, and payment types.

Real-Time Monitoring

Visualizing key performance indicators (KPIs) to flag at-risk orders before they miss delivery deadlines.

Interactive Dashboard Interface:
Visualizing Delivery Reliability & Seller Performance Metrics

Operational Impact

Moving from retrospective analysis to proactive intervention through data-driven evidence.

Key Findings Summary

Finding Area	Key Insight
Top Late Category	**Bed Table Bath** (739 late orders) is the worst-performing product category for on-time delivery.
Top Late Seller	Seller 7c67e1448b had **200 late deliveries** — nearly 2x the second-highest seller.
Payment Bottleneck	UPI payments average **1.77 days** to approval vs 0.19 days for credit card — a 9x gap.
Weight & Freight	Heavier products (5,682g avg) take **~15.2 days** to deliver vs 5.7 days for lightweight items.
The Trust Gap	Estimated delivery dates are frequently **2x the actual delivery time**, reducing conversion rates.
Late Shipping Risk	Health Beauty and Bed Table Bath dominate both late delivery AND late shipping categories.

Strategic Recommendations

Seller Performance

Implement **SLA enforcement** and performance improvement plans for chronically late sellers. Use financial penalties for persistent lateness to drive accountability.

Payment Optimization

Address the **9x gap** in UPI approval latency by optimizing banking API integrations or pre-clearing orders to accelerate the fulfillment pipeline.

ML-Based Estimator

Build a **weight-aware delivery estimator**. Incorporating product weight and freight value will provide more accurate dates, increasing conversion without over-promising.

Logistics Partnerships

Establish **category-specific carrier partnerships** for bulky items like "Bed Table Bath" to ensure specialized handling and faster transit times.

"The Trust Gap is a Leaky Bucket."

Every late delivery costs not just one order — it costs future orders from the same customer. Fixing delivery reliability is the highest-ROI investment an e-commerce platform can make.

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